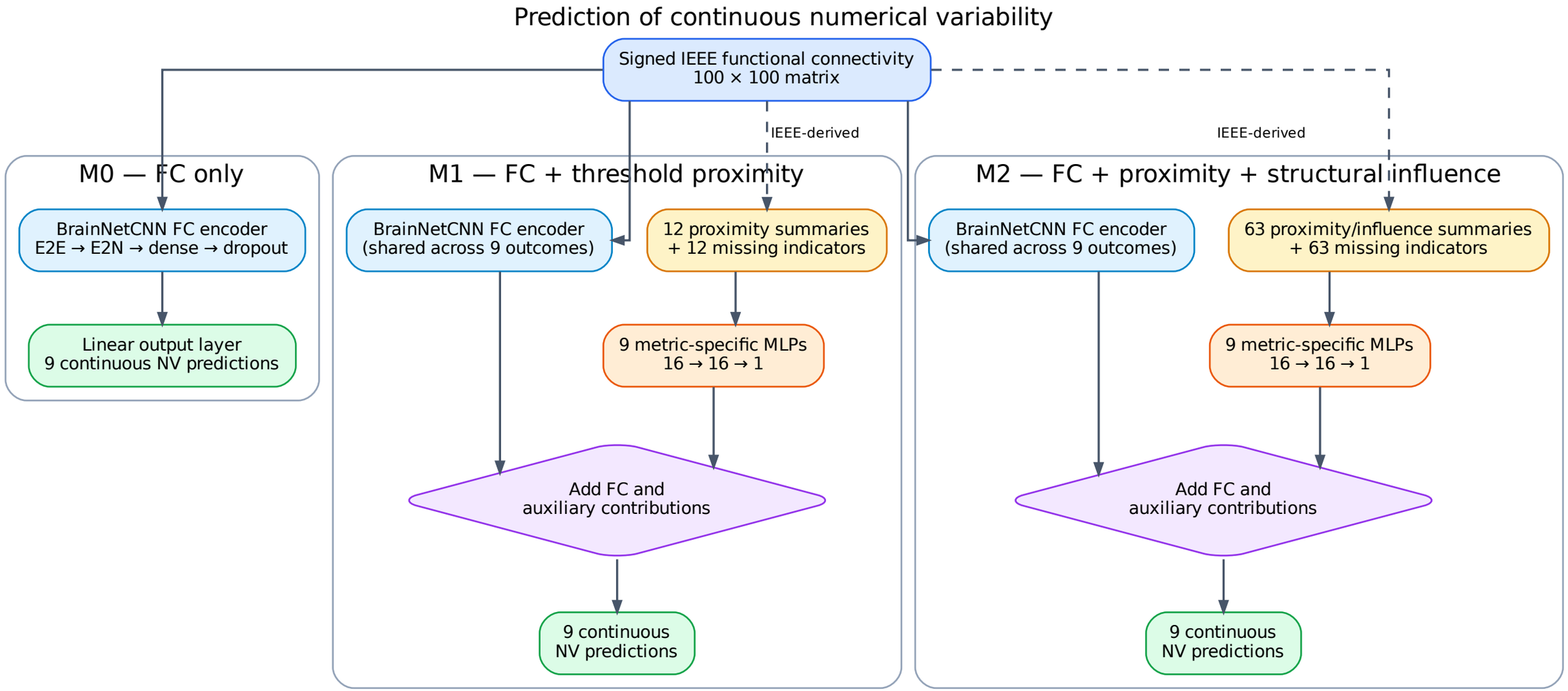
Prediction of continuous numerical variability
Signed IEEE functional connectivity
100 × 100 matrix
IEEE-derived
IEEE-derived
M0 — FC only
BrainNetCNN FC encoder
E2E → E2N → dense → dropout
Linear output layer
9 continuous NV predictions
M1 — FC + threshold proximity
BrainNetCNN FC encoder
(shared across 9 outcomes)
12 proximity summaries
+ 12 missing indicators
9 metric-specific MLPs
16 → 16 → 1
Add FC and
auxiliary contributions
9 continuous
NV predictions
M2 — FC + proximity + structural influence
BrainNetCNN FC encoder
(shared across 9 outcomes)
63 proximity/influence summaries
+ 63 missing indicators
9 metric-specific MLPs
16 → 16 → 1
Add FC and
auxiliary contributions
9 continuous
NV predictions